

SMART INDIA HACKATHON 2026

# SIH26023

## **AI-Powered Geological, Mining and other Reporting Solution for CMPDI/CIL subsidiaries**

Deep Research • Feasibility • Architecture • Competitive Landscape • Implementation Plan

Prepared for team-level problem statement evaluation

Based on the supplied SIH26023 problem statement pages and external research current to 22 September 2026

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# 1. Executive Summary

SIH26023 asks for an AI-assisted reporting platform for CMPDI/CIL subsidiaries that can turn heterogeneous historical and contemporary records into reliable reports, analytics and fast answers to high-priority or parliamentary-style questions. The supplied problem statement specifically identifies scanned PDFs, digital documents, spreadsheets, images and historical archives as inputs, and calls for three visible capabilities: automated report generation, automated word-cloud/topic identification, and an AI-based query-and-response system.

## Core interpretation

The strongest implementation is not a generic PDF chatbot. It should be a traceable data-to-report pipeline: ingest → digitize → extract → validate → reconcile → retrieve → answer → generate deliverable, with every important number linked back to its source document, page and extraction context.

## Research findings

- CMPDI already operates multiple information systems, including the Online Coal Block Information System (OCBIS), Mine Data Management System (MDMS), National Coal Portal and other workflow portals. The proposed system should therefore sit as an intelligence/reporting layer and connector rather than attempt to replace every existing system. [1]
- CMPDI reports are genuinely heterogeneous and high-volume: its 2024-25 annual report describes geological, hydrogeological, environmental, project and other technical reporting work, including dozens of report outputs and large-scale exploration activity. [2]
- Public sources are sufficient to build a realistic demonstration corpus: CMPDI annual reports, Ministry of Coal annual reports, Coal India public reporting pages and OCBIS provide examples of production, project, geological and block-level information. [1][2][3][4]
- The baseline technology stack is practical on student hardware: OCR/table extraction can run largely on CPU; a 6 GB GPU is useful but not mandatory; a compact quantized vision/reasoning model can handle selected document pages while structured extraction and analytics remain deterministic.
- The main differentiation challenge is that public SIH26023 prototypes already exist. Several current repositories implement the obvious pattern of document ingestion, structured fact extraction, evidence-backed retrieval and automated report composition. [5][6]

## Overall feasibility

High for a student software team. The harder part is not making a UI or calling an LLM; it is building reliable numeric extraction, period/unit normalization, conflict handling, evidence citations and repeatable report generation.

# 2. Problem Statement at a Glance

| Field                | Value                                                                                 |
|----------------------|---------------------------------------------------------------------------------------|
| Problem Statement ID | SIH26023                                                                              |
| Title                | AI-Powered Geological, Mining and other Reporting Solution for CMPDI/CIL subsidiaries |
| Organization         | Ministry of Coal                                                                      |

|                                |                                                                                                                                                |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Department                     | Coal India Limited                                                                                                                             |
| Category                       | Software                                                                                                                                       |
| Primary scope from supplied PS | AI-assisted document processing, reporting, topic/word-cloud analysis, and AI query/response for geological, mining and production information |

**Source/Note:** Problem statement title, organization, category, background, objectives and requested modules are transcribed from the three pages supplied in this conversation. External pages use [1]-[6] below.

## Explicit requirements from the supplied PS

- Automate processing of scanned PDFs, digital documents, spreadsheets, images and historical archives.
- Improve validation, consistency and traceability across historical and contemporary datasets.
- Provide an automated report generation platform.
- Provide an automated word-cloud and topic-identification module.
- Provide an AI-based query-and-response system.
- Quantify report-time reduction, structured-extraction/report-generation accuracy and automation coverage.
- Support a phased rollout from data digitization and preprocessing through platform development, testing, workflow integration, training and continuous enhancement.

## 3. What the Problem Really Requires

The phrase “AI-powered reporting” can hide several different engineering problems. For this PS, the most important distinction is between generative text and trustworthy structured information. The system has to answer questions about figures, periods, subsidiaries and reports. That makes deterministic extraction, provenance and reconciliation first-class components.

| Requirement        | What a weak solution does    | What a strong solution should do                                            |
|--------------------|------------------------------|-----------------------------------------------------------------------------|
| Scanned PDFs       | OCR text only                | OCR + page/layout coordinates + table structure + confidence                |
| Numeric extraction | Copies numbers into text     | Stores entity, metric, value, unit, period, source and evidence pointer     |
| Historical data    | Vector search only           | Normalizes fiscal years, subsidiaries, units and revisions                  |
| Conflicts          | Chooses one value silently   | Surfaces conflicting values with receipts and possible causes               |
| AI answers         | Free-form generated response | Grounded response with exact source/page/table citations                    |
| Reports            | LLM-generated prose          | Template-driven documents built from validated facts and evidence           |
| Topics             | Simple word frequency        | Topic clusters + keyphrases + time/entity filtering                         |
| Evaluation         | Demo screenshots             | Gold-set extraction accuracy + answer citation accuracy + end-to-end timing |

## 4. CMPDI/CIL Operating Context and Existing Digital Landscape

External research indicates that the surrounding ecosystem is already digitally active. CMPDI describes itself as providing geological exploration, project planning/design, engineering, environmental and geomatics services, including remote sensing, GIS, GPS, digital photogrammetry, LiDAR and UAV-related work. [2][7]

### Design implication

The new platform should not be positioned as a replacement for CMPDI's existing operational portals. It should create a common intelligence layer that can ingest or connect to those systems, reconcile records, and generate human-readable reporting outputs.

### Existing systems / integration points

| Existing capability        | What it appears to provide                                                                   | Why SIH26023 should care               | Integration idea                           |
|----------------------------|----------------------------------------------------------------------------------------------|----------------------------------------|--------------------------------------------|
| OCBIS                      | Online coal block information; block/area/resource data                                      | Geological/block context               | Read/export connectors; preserve block IDs |
| MDMS                       | Project monitoring incl. clearances, land, R&R, financial/production and infrastructure data | Contemporary project context           | API/database/file connector + snapshots    |
| National Coal Portal       | Dashboard-level coal information                                                             | High-level production / policy context | Read-only connector or curated import      |
| Document Management System | Central document access                                                                      | Historical source corpus               | Controlled document ingestion              |
| e-Office support           | Organization-wide office workflow                                                            | Potential approval / document workflow | Exportable report packages + metadata      |

**Source/Note:** CMPDI 2024-25 Annual Report lists OCBIS, MDMS, National Coal Portal, e-Office support and other portals. [2]

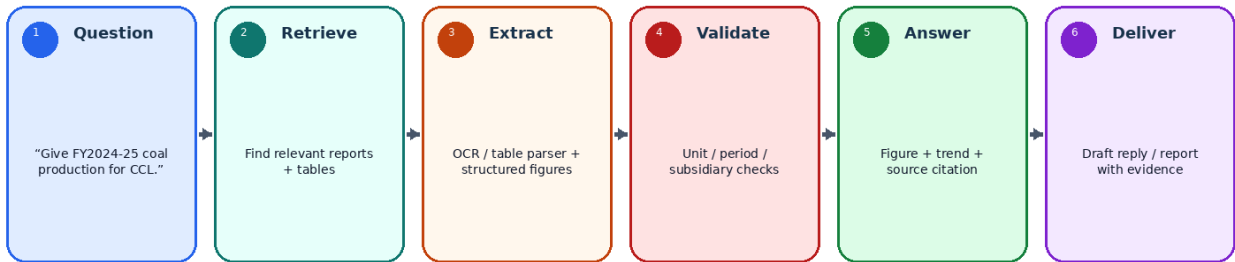
### Why this matters for the demo

- Show that the system is complementary to existing government/CIL digital infrastructure.
- Use public-source replicas or exported examples rather than pretending to access proprietary databases.
- Demonstrate connector boundaries: source remains authoritative; the AI layer adds search, validation, reporting and traceability.

## 5. Target Users and Core Workflows

| User / stakeholder                   | Typical need                                     | System response                                                 |
|--------------------------------------|--------------------------------------------------|-----------------------------------------------------------------|
| Reporting officer                    | Prepare recurring production/geological reports  | Generate draft from validated data + template                   |
| Geologist / technical analyst        | Find historical resource/exploration information | Evidence-backed search + comparisons                            |
| Management / Ministry-facing officer | Respond to high-priority inquiries               | Fast natural-language query + exact source                      |
| Data/IT administrator                | Ingest archives and monitor quality              | Pipelines, provenance, validation and exceptions                |
| Reviewer / approver                  | Check generated output before release            | Side-by-side evidence, conflicts, confidence and approval state |

Illustrative Parliamentary / Administrative Query Workflow



Key principle: every numeric answer must retain a verifiable path back to document → page → table/cell or text span.

Source/Note: Workflow diagram is a proposed design, not an existing CMPDI process map.

## 6. Proposed Solution: Mining Reporting Intelligence Platform

Working concept name: MRIP — Mining Reporting Intelligence Platform. The name is only a project label; the proposed design is what matters.

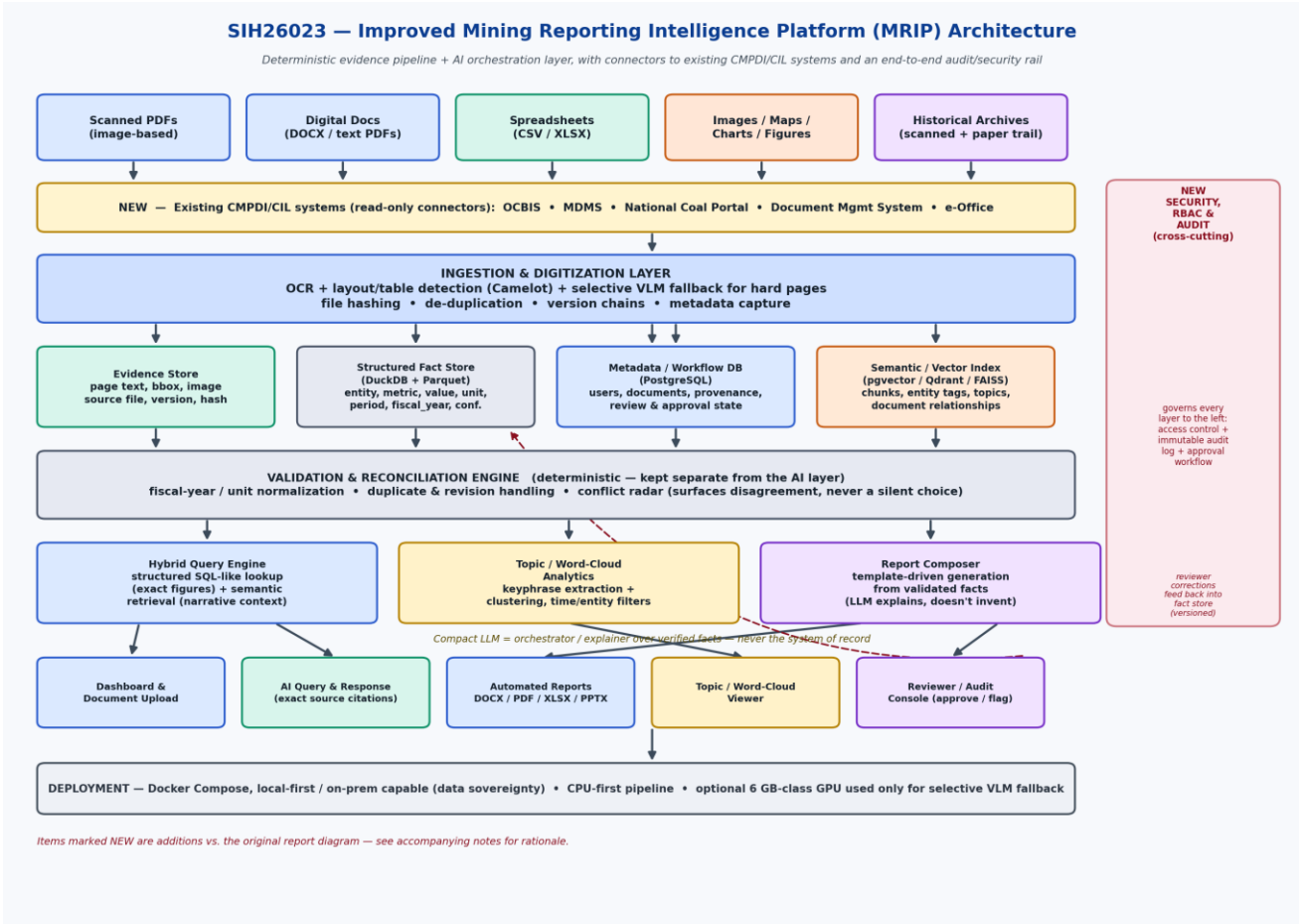
MRIP is a local-first document intelligence and reporting system that creates a persistent evidence layer between raw files and final reports. Its central design principle is that generated answers and reports should be constructed from validated structured facts and traceable source evidence, not from unverified LLM text.

### Core modules

| # | Module                             | Purpose                                                                                                  |
|---|------------------------------------|----------------------------------------------------------------------------------------------------------|
| 1 | Ingestion & digitization           | PDF/DOCX/XLSX/image intake, OCR, table extraction, file hashing, metadata and versioning                 |
| 2 | Evidence store                     | Page-level text, table/cell references, bounding boxes, source file, version, confidence                 |
| 3 | Structured figure store            | Production, dispatch, resources, drilling, project and other numeric facts with normalized units/periods |
| 4 | Knowledge index                    | Semantic chunks, entity tags, topics and document relationships                                          |
| 5 | Validation / reconciliation engine | Unit checks, date/fiscal-year checks, duplicate/revision handling and conflict radar                     |

|   |                            |                                                                                                                   |
|---|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| 6 | Report generation          | Templates for production summary, comparative analysis, administrative/parliamentary response and technical brief |
| 7 | Topic/word-cloud analytics | Keyphrases, topic clusters and trends by time/entity                                                              |
| 8 | AI query engine            | Natural-language interface that combines structured SQL-like retrieval with evidence-grounded semantic search     |
| 9 | Review and audit           | Confidence, evidence panel, reviewer status, generated-file provenance                                            |

7. Technical Architecture



Source/Note: Architecture is the proposed reference design for the hackathon MVP.

Why a hybrid architecture is preferable

- Exact figures should come from a structured store or deterministic extraction layer whenever possible.
- Narrative context and historical correspondence benefit from semantic retrieval.
- An LLM should orchestrate and explain evidence rather than become the database of record.
- Report generation should use validated fields and templates so the same input produces consistent output structure.

## 8. Document Digitization and Data Engineering

The PS is unusual in explicitly mixing scanned PDFs, office files, spreadsheets, images and archives. Treat them as different document classes with different extraction pipelines.

| Input              | Primary method         | Fallback / QA                           | Stored output                       |
|--------------------|------------------------|-----------------------------------------|-------------------------------------|
| Text PDF           | Text-layer parsing     | Render + OCR when text quality is poor  | Text spans + page IDs               |
| Scanned PDF        | OCR + layout detection | Vision model for hard pages             | OCR spans + boxes + page images     |
| Digital XLSX/CSV   | Direct parser          | Schema/units validation                 | Typed rows + source cell            |
| Tables in PDF      | Camelot/other parser   | Image/table model + manual review queue | Table cells + extraction confidence |
| Images / charts    | Vision/OCR             | Human verification for ambiguous values | Objects/text + evidence             |
| Historical archive | Batch pipeline         | Checksum + dedup + version chains       | Canonical document + metadata       |

**Source/Note:** Camelot supports multiple table parsing methods and exposes parsing metrics; importantly, it is designed for text-based PDFs, so scanned documents need an OCR/vision stage first. [8]

### Proposed canonical fact schema

```
Fact {
  entity_id, subsidiary, mine_or_block,
  metric, value, unit, period_start, period_end,
  fiscal_year, source_document, source_version,
  page, table_id, cell_or_bbox,
  extraction_method, confidence, status
}
```

## 9. AI / ML Strategy

The system should use AI where ambiguity or language understanding is genuinely helpful, and deterministic code where correctness is more important.

| Task                       | Recommended approach                            | Reason                                      |
|----------------------------|-------------------------------------------------|---------------------------------------------|
| OCR on scanned reports     | OCR engine + layout model                       | Deterministic, inspectable, page-aware      |
| Difficult tables / figures | Vision-language model as fallback               | Useful on irregular layouts/images          |
| Topic identification       | Keyphrase extraction + BERTopic-like clustering | More analytical than simple word count      |
| Numeric question           | Structured query over fact store                | Reduces hallucination risk                  |
| Narrative question         | Hybrid retrieval + compact LLM                  | Combines context with source evidence       |
| Report drafting            | Template + controlled generation                | Consistent structure and predictable output |
| Anomaly / conflicts        | Rules + statistical checks                      | Explicit and testable                       |

**Model strategy**  
A compact vision-language model can be kept as a selective fallback rather than used for every page. This reduces GPU cost and improves reproducibility. Qwen3-VL-2B-Instruct is one current open-weight multimodal

option and is Apache-2.0 licensed. [9]

## 10. Automated Report Generation

The report generator should behave more like a publishing pipeline than a chatbot. Each report template defines required fields, calculations, charts, evidence blocks and approval metadata.

| Report type                          | Typical contents                                    | Automation design                             |
|--------------------------------------|-----------------------------------------------------|-----------------------------------------------|
| Production summary                   | Target vs actual, YTD, growth, subsidiary breakdown | Structured query + chart + template           |
| Comparative analysis                 | Year-over-year / subsidiary comparison              | Normalized facts + deterministic calculations |
| Geological / exploration brief       | Resource, drilling, block information               | Evidence retrieval + structured tables        |
| Administrative / parliamentary draft | Question, figure, contextual explanation, sources   | Query engine + cited evidence + template      |
| Technical note                       | Selected topic, findings, references                | Hybrid retrieval + human review               |

### Key rule

Never allow the LLM to invent the underlying number. The report engine should fetch validated facts, then ask the language model only for explanation, transitions or controlled narrative where appropriate.

## 11. Word Cloud and Topic Identification

The word-cloud requirement is straightforward, but the useful version is richer: combine a visual word cloud with keyphrases, topic clusters and filters by year, subsidiary, report type or geological topic.

- Baseline: TF-IDF / noun-phrase frequencies.
- Better: BERTopic-like embeddings + clustering for thematic groups.
- Useful UI: click a topic to show documents, pages and representative passages.
- Longitudinal view: track topic prevalence over fiscal years or reporting periods.

### Important

The word cloud satisfies an explicit PS requirement, but it should not be the centerpiece of the demo. The high-value feature is the link from topic → evidence → documents → report/query response.

## 12. AI Query & Response with Evidence

A good query engine should classify each question before answering it. The system can route numeric questions to structured data and narrative questions to hybrid retrieval.

| Question type           | Route                        | Output                                | Guardrail                          |
|-------------------------|------------------------------|---------------------------------------|------------------------------------|
| Exact production figure | Structured fact store        | Number + period + subsidiary + source | No answer if evidence is ambiguous |
| Trend / comparison      | Structured query + analytics | Table/chart + calculation             | Unit / period normalization        |

|                                |                                |                                |                                            |
|--------------------------------|--------------------------------|--------------------------------|--------------------------------------------|
| Why did a value change?        | Hybrid retrieval               | Narrative + supporting records | Cite evidence and flag uncertainty         |
| Find all documents about topic | Semantic search + topic index  | Document/page list             | Ranked results with snippets               |
| Draft parliamentary reply      | Structured + hybrid + template | Draft response document        | Mandatory source citations + review status |

## 13. Data Validation, Reconciliation and Traceability

This is the core area where SIH26023 can become more than a document chatbot. Mining reports often contain revised numbers, different periods, different units and multiple views of the same metric. The application should make those differences visible.

### Unit normalization

Normalize MT, lakh tonnes, crore, %, m, km, ha and other common units into canonical forms while retaining the original representation.

### Fiscal-year normalization

Store FY2024-25 explicitly as a date range and never mix it silently with calendar-year data.

### Versioning

Keep the newest document as current but preserve superseded versions for audit and comparison.

### Conflict radar

If two documents give different figures for the same entity/metric/period, show them side-by-side and require a resolution state.

### Provenance

Every displayed value should be traceable to document, page, table/cell or bounding box.

### Confidence

Separate OCR confidence, parser confidence and semantic-answer confidence. Do not collapse them into one unexplained number.

## 14. Data Sources and Demonstration Corpus

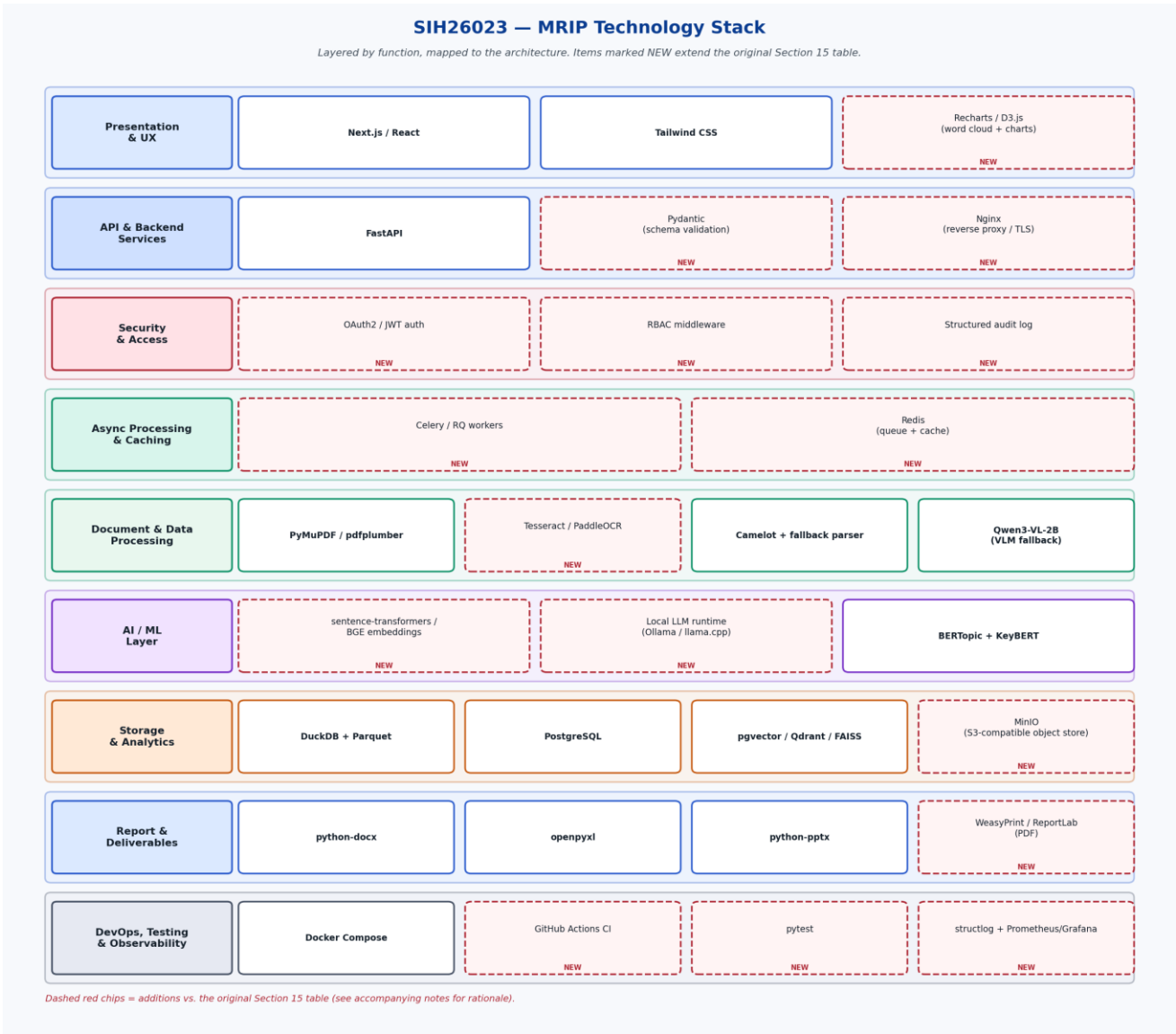
No proprietary dataset is needed to prove the concept. The strongest demo corpus can be built from public government/company material and clearly labeled synthetic scenarios.

| Source                          | Examples                                 | Use in prototype                                 | Caveat                             |
|---------------------------------|------------------------------------------|--------------------------------------------------|------------------------------------|
| CMPDI annual reports            | 2021-22 through 2024-25                  | Geological / hydrogeological / project reporting | Large PDFs; mixed structure        |
| Ministry of Coal annual reports | 2024-25 sections on exploration/projects | Policy and sector context                        | Long narrative reports             |
| Coal India production pages     | Production / offtake summaries           | Structured/current-style numeric queries         | Some pages require authentication  |
| OCBIS                           | Coal block / resource information        | Block and geological context                     | Data is provisional and can change |

|                                 |                                  |                            |                            |
|---------------------------------|----------------------------------|----------------------------|----------------------------|
| Public PDF tables               | Selected reports with tables     | Table extraction benchmark | Need gold labels           |
| Synthetic parliamentary queries | Controlled question set          | End-to-end QA benchmark    | Must be labeled synthetic  |
| Synthetic conflicting documents | Two versions with changed figure | Conflict radar demo        | Clearly label as test data |

**Source/Note:** CMPDI annual reports and OCBIS are public. Coal India exposes production dashboards/pages with production and offtake information. [1][2][3][4]

15. Technology Stack



**Source/Note:** Layered technology-stack view mapped to the architecture; the table below gives the same topic choices with reasons.

| Layer    | Suggested choice           | Reason                              |
|----------|----------------------------|-------------------------------------|
| Frontend | Next.js / React + Tailwind | Fast internal dashboard development |
| API      | FastAPI                    | Python ecosystem for OCR/data/AI    |

|                             |                                            |                                                              |
|-----------------------------|--------------------------------------------|--------------------------------------------------------------|
| Document processing         | PyMuPDF / pdfplumber + OCR engine          | Text/page extraction                                         |
| Table extraction            | Camelot + fallback parser                  | Text-PDF table extraction with measurable parsing output [8] |
| Structured analytics        | DuckDB + Parquet                           | Fast local analytical queries across CSV/Parquet/JSON [10]   |
| Metadata / transactional DB | PostgreSQL                                 | Users, documents, provenance, workflow state                 |
| Semantic retrieval          | pgvector or Qdrant/FAISS                   | Local embeddings + vector search                             |
| Models                      | Compact VLM + text model via local runtime | Keep inference local and controllable                        |
| Report generation           | python-docx + openpyxl + python-pptx       | Actual Word/Excel/PPT deliverables                           |
| Packaging                   | Docker Compose                             | Repeatable local deployment                                  |

## 16. Hardware / Compute Feasibility

SIH26023 is software. The main compute question is how much local AI inference is needed. Most of the pipeline—PDF parsing, OCR, tables, DuckDB, database, retrieval and report generation—can run on CPU. GPU is mainly useful for vision-language inference and larger language models.

| Hardware level      | Practical setup                     | What it supports                                       | Assessment                                                   |
|---------------------|-------------------------------------|--------------------------------------------------------|--------------------------------------------------------------|
| Basic               | 16 GB RAM, modern 6–8 core CPU, SSD | OCR + extraction + retrieval + small local model       | Sufficient for core MVP                                      |
| Recommended student | 32 GB RAM + 6–8 GB GPU              | Faster VLM pages + local reasoning + all core services | Good balance                                                 |
| Strong demo         | 32–64 GB RAM + 12–16 GB GPU         | Larger models, faster multimodal batches               | Comfortable                                                  |
| Server              | 64–128 GB RAM + 24 GB+ GPU          | Parallel inference / multiple models                   | Useful for expanded deployment                               |
| 6 GB RTX 3050       | 6 GB VRAM + sufficient system RAM   | Small/quantized VLM + OCR/CPU pipeline                 | Sufficient for this PS; design around selective AI inference |

### Compared with SIH26117

SIH26023 is significantly less GPU-dependent. A 6 GB RTX 3050 is adequate for a prototype because OCR/table extraction and most numeric analytics can remain CPU/deterministic, while a compact vision model can handle selected difficult pages.

## 17. Evaluation Framework and Measurable KPIs

The supplied PS explicitly asks benefits to be quantified. The team should therefore define a benchmark before optimizing the UI.

| KPI                            | Definition                              | Suggested MVP target              | How to measure                 |
|--------------------------------|-----------------------------------------|-----------------------------------|--------------------------------|
| Structured extraction accuracy | Correct numeric facts / gold facts      | ≥ 95% on curated gold set         | Cell/fact-level evaluation     |
| Source attribution accuracy    | Answers pointing to correct source/page | ≥ 95%                             | Human-labelled query benchmark |
| Report generation automation   | Report sections completed automatically | ≥ 80% of selected template        | Section checklist              |
| Report preparation time        | Human baseline vs system                | Demonstrate substantial reduction | Timed repeated runs            |

|                           |                                    |                                     |                                  |
|---------------------------|------------------------------------|-------------------------------------|----------------------------------|
| OCR high-confidence share | Pages/fields above threshold       | Track by document type              | OCR metadata                     |
| Conflict detection recall | Known conflicts detected           | High on curated tests               | Injected revision/conflict cases |
| Query latency             | Median answer time                 | < 10 s for local structured queries | Application logs                 |
| Manual-review rate        | Fraction of facts requiring review | Reduce over iterations              | Workflow logs                    |

## 18. Implementation Roadmap

| Phase                       | Deliverable                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------|
| Phase 0 — Corpus & gold set | Collect 20–50 representative public documents; create ground-truth facts and 25–50 benchmark queries. |
| Phase 1 — Ingestion         | PDF/DOCX/XLSX/image upload, OCR, table extraction, metadata, hashing and document registry.           |
| Phase 2 — Structured store  | Normalize entities, metrics, periods and units; store evidence pointers.                              |
| Phase 3 — Search & QA       | Hybrid retrieval, structured query engine, evidence panel and citations.                              |
| Phase 4 — Reports           | DOCX/Excel templates, charts, report package and reviewer workflow.                                   |
| Phase 5 — Topics            | Word cloud, keyphrases, topic clusters and trend filters.                                             |
| Phase 6 — Hardening         | Conflict radar, confidence, regression tests, performance tuning, deployment packaging.               |

## 19. Hackathon Demo Plan

A good demo should tell one complete story instead of showing ten disconnected features.

1. Upload a scanned historical report and a current spreadsheet.
2. System OCRs the report, extracts tables and creates structured facts with source references.
3. Ask: “Compare production for Subsidiary A across FY2023-24 and FY2024-25.”
4. System answers from the structured fact store and opens the exact evidence pages.
5. Ask a narrative question about the cause/context of a change; hybrid retrieval brings relevant paragraphs and reports.
6. Generate a production-summary DOCX with table, chart, executive summary and sources appendix.
7. Open Topic Intelligence to show key themes and a word cloud, filtered by year/document type.
8. Show Conflict Radar on a synthetic revised report and explain why the system did not silently merge conflicting numbers.

### Strongest demo moment

Clicking a number in the final report should open the underlying document page/table evidence. This turns “AI generated report” into “AI-assisted, traceable report.”

## 20. Risks and Mitigations

| Risk                | Impact              | Mitigation                                             |
|---------------------|---------------------|--------------------------------------------------------|
| OCR errors in scans | Wrong numeric facts | Confidence thresholds + visual evidence + review queue |

|                                       |                     |                                                                                  |
|---------------------------------------|---------------------|----------------------------------------------------------------------------------|
| LLM hallucination                     | Incorrect answers   | Structured route for numbers + citations + no-answer state                       |
| Conflicting versions                  | Wrong report value  | Version chains + conflict radar                                                  |
| Poor table extraction                 | Incomplete datasets | Multiple parsers + gold-set regression tests                                     |
| Model too slow on venue hardware      | Demo failure        | Small model + batch preprocessing + CPU-first pipeline                           |
| Public data mismatch to real workflow | Weak domain fit     | Use the PS workflow terminology and build template-driven outputs                |
| Overbuilding word cloud               | Time lost           | Keep topic module compact; prioritize core reporting path                        |
| Integration assumptions               | Misleading claims   | Label public/demo connectors separately from hypothetical enterprise integration |

## 21. Competitive Landscape and Differentiation

Public GitHub activity around SIH26023 shows that the baseline concept is already being implemented by multiple teams. Examples include repositories describing heterogeneous document upload, structured fact extraction, evidence-backed retrieval, conflict detection and automated report generation. [5][6][11]

| Observed baseline     | Already appearing publicly | Where to differentiate                    | Priority |
|-----------------------|----------------------------|-------------------------------------------|----------|
| Document upload + OCR | Yes                        | Source-linked evidence + quality scoring  | High     |
| RAG / semantic search | Yes                        | Hybrid structured + semantic retrieval    | High     |
| Report generation     | Yes                        | Template integrity + fact-level receipts  | High     |
| Conflict handling     | Appears in some projects   | Formal revision graph + reviewer workflow | High     |
| Topic / word cloud    | Expected by PS             | Interactive topic-to-evidence exploration | Medium   |
| Generic chatbot UI    | Common                     | Do not compete on UI alone                | Low      |

### Differentiation thesis

Make the system an evidence and reporting engine, not a mining-themed chatbot: fact-level provenance, version-aware reconciliation, deterministic numeric answers, human approval states and reproducible report packages.

## 22. Comparison with SIH26117

SIH26117 (MRPL) and SIH26023 both involve confidential industrial knowledge work and local AI, but they emphasize different systems. SIH26117 is a general sovereign agentic workbench with multimodal models, model routing, local tools, code execution and air-gap proof. SIH26023 is more domain-specific: document digitization, mining/production/geological reporting, topic analysis and evidence-backed query/response.

| Dimension     | SIH26023                           | SIH26117                             |
|---------------|------------------------------------|--------------------------------------|
| Primary value | Reliable mining/reporting workflow | General sovereign AI workbench       |
| Data types    | PDF, spreadsheet, image, archive   | Broader multimodal industrial inputs |

|                      |                                         |                                                |
|----------------------|-----------------------------------------|------------------------------------------------|
| GPU dependence       | Low–medium                              | Medium–high                                    |
| Core difficulty      | Extraction + normalization + provenance | Agent orchestration + multimodal + sovereignty |
| Key demo             | Question → evidence → report            | Agent → tools → deliverable + air-gap proof    |
| Domain specificity   | High to coal/CMPDI/CIL reporting        | Moderate; designed for multiple industries     |
| Competition signal   | Multiple public SIH26023 prototypes     | Many public SIH26117 prototypes                |
| Best differentiation | Traceable numbers + reconciliation      | Workflow depth + security + agent quality      |

**Source/Note:** SIH26117 requirements are taken from the user-supplied SIH26117 problem statement file and current public repositories. Examples of public SIH26117 prototypes are cited in [12]–[15].

## 23. Team / Delivery Plan

| Role                   | Responsibilities                             | Main output                     |
|------------------------|----------------------------------------------|---------------------------------|
| Document/data engineer | OCR, extraction, schemas, validation         | Fact store + ingestion pipeline |
| AI/ML engineer         | Embeddings, VLM fallback, retrieval, topics  | Models + evaluation             |
| Backend engineer       | FastAPI, workflow, database, reports         | Core services                   |
| Frontend engineer      | Upload, evidence viewer, query UI, analytics | Dashboard                       |
| QA/domain analyst      | Gold labels, benchmark, domain scenarios     | Test set + KPI results          |
| Pitch/demo owner       | Narrative, screenshots, demo sequencing      | Judging presentation            |

## 24. Feasibility Assessment

| Area                            | Assessment                                | Reason                                                                |
|---------------------------------|-------------------------------------------|-----------------------------------------------------------------------|
| Software feasibility            | High                                      | All major building blocks have mature open-source options             |
| Data availability for prototype | High                                      | Public annual reports and portals provide realistic documents         |
| Hardware cost                   | Low–medium                                | No special hardware is required                                       |
| 6 GB GPU suitability            | Adequate                                  | Use compact/quantized VLM selectively                                 |
| AI complexity                   | Medium                                    | Core value comes from hybrid retrieval + extraction, not giant models |
| Domain complexity               | High                                      | Mining terminology, periods and metrics require careful schema design |
| Demo reliability                | High if deterministic core is prioritized | Avoid generative dependence for numeric facts                         |
| Differentiation difficulty      | High                                      | Public prototypes already cover baseline features                     |

## 25. Recommended MVP Scope

For a hackathon, the team should resist implementing every possible mining use case. A coherent MVP can satisfy the PS with a narrow but deep workflow.

- Corpus: 20–50 public CMPDI/CIL/Ministry documents + 5–10 spreadsheets.
- Extraction: OCR + table extraction + structured fact store.
- Validation: fiscal year, units, subsidiary/entity normalization, duplicate/version checks.
- Query: 25–50 gold-standard questions, including exact-figure and comparison queries.
- Reports: 2 polished templates — Production Summary and Parliamentary/Administrative Response.
- Topics: word cloud + topic list + click-through to source pages.
- Evidence: page-level citation and fact-level source receipt for every answer/report number.
- Conflict demo: one synthetic revised document pair showing safe handling of disagreement.
- Optional: connector abstraction for OCBIS/MDMS/CIL portals, demonstrated with a mock adapter rather than claimed live access.

## 26. Conclusion

SIH26023 is a practical, domain-specific AI problem with a strong prototype path. Its real engineering value lies in converting messy source material into structured, validated and traceable information, then using that information to answer questions and generate repeatable reports. Existing CMPDI/CIL digital systems make an integration-layer strategy more credible than a replacement strategy. [2]

The report recommends treating the LLM as one component of a broader evidence system. OCR, table extraction, schemas, fiscal-year/unit normalization, conflict handling and provenance should remain explicit software components. This architecture also reduces dependence on expensive GPUs and makes an RTX 3050 6 GB class development machine viable for a prototype.

### Bottom line

A strong SIH26023 submission should feel like a small internal government reporting product—not a generic chatbot. The defining feature should be that every important answer or number can be verified back to the exact source evidence, while the platform automates the repetitive path from archived document to reviewed report.

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